

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I M. Pavankalyan (192312178) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



CSA17

Assessment Tool 2

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Sub: Artificial Intelligence.

- ① A University must assign 3 Professors (P_1, P_2, P_3) to 3 courses (AI, Data Structures, Networks) offered in 3 different time slots such that:

Constraints:

- ① P_1 cannot teach in the network slot.
 - ② P_2 must be scheduled before P_3 (AI < Data Structures < Networks)
 - ③ only one professor per course/slot
 - ④ P_3 cannot teach AI.
 - ⑤ Every professor must be assigned
- Apply backtracking of the system.
State the final valid schedule.

Backtracking Search for University.

A University must assign 3 professors (P_1, P_2, P_3) to 3 courses offered in 3 time different slots.

Courses and Time Slots

Time Slot

slot 1

slot 2

slot 3

Professors:

P_1

P_2

P_3

Course

AI

Data Structure

Networks

Constraints:

1. P_1 cannot teach in the network slots.
2. P_2 must be scheduled before P_3 .

order of slots.

AI < Data Structure.

3. only one professor can be assigned to each course.

4. P_3 cannot teach AI

5. Each professor must teach exactly one course.

Step 1: Formulating the CSP

Variables:

AI

Data Structures.

Networks

Each Variable represents the professor

Domain:

Each course can be assigned one of the professors.

$$A1 = (P_1, P_2, P_3)$$

$$\text{Data Structures} = \{P_1, P_2, P_3\}$$

$$\text{Networks} = \{P_1, P_2, P_3\}$$

Step 2:

Apply constraints:

P_1 cannot teach Networks

Therefore,

$$\text{Networks} \neq P_1$$

Remaining possibilities.

$$\text{Networks} = \{P_2, P_3\}$$

Constraint 2:

P_3 cannot teach A.I.

Therefore.

$$A1 \neq P_3$$

Remaining possibilities.

$$A1 = \{P_1, P_2\}$$

Constraint 3:

Each Professor teaches exactly one course.

No professor can appear twice.

Constraint 4:-

P_2 must teach before P_3

Possible valid orders are.

$P_2 \rightarrow AI, P_3 \rightarrow \text{Data Structures}$.

$P_2 \rightarrow AI, P_3 \rightarrow \text{Networks}$

Invalid orders include:-

P_3 before P_2

P_2 in Networks while P_3 is in AI or Data Structure.

Solution found:-

Course

AI

Data Structures

Networks

Professor.

P_1

P_2

P_3

Since every constraint is satisfied.

cleaning robot operates in a 5×5 room-grid from Start (S) to a docking Goal (G) with a limited battery allowing at most 8 moves.

Constraints:

- (i) Movement allowed: up, Down, Left, Right.
- (ii) Cost of each move = 1
- (iii) Cannot pass through furniture.
- (iv) The robot must reach G within the battery limit.

Using the above constraints and depth limited searches.

Problem Statement:

Cleaning robot operates in a 5×5 room grid. The robot starts from the start (S) position and must reach the docking station before its battery runs out. The robot cannot move through obstacle cells.

task is to apply the depth limited search algorithm with the battery limit as the depth limit

Show the step by step node expansion and determine whether the robot can reach the goal within the show step by step node expansion, and determine whether the robot can reach the goal within the battery limit.

Given constraints:

Grid Size = $s \times s$

Start position = S

Goal position = G

Maximum battery = 8 moves.

Allowed movements:

- UP (U)

- Down (D)

- Left (L)

- Right (R)

- Each move costs one battery unit.

- Robot cannot move through the furniture.

Room Grid Representation:

S	.	X	.	.
.		X	.	.
.		.	.	.

Legend:-

S = Start

G = Goal

X = obstacle

 = free cell

Step 1:-

Robot starts at

(1,1)

Depth = 0

Battery used = 0

Remaining Battery = 8

Apply depth:

Depth Limit = 8

The robot expands one node at a time while keeping the search depth or equal to 8.

Node expansion:-

$S(1,1)$

Depth = 0

Possible moves

- Right
- Down

choose Right first.

Right is obstacle.

choose down.

Level 3:-

Robot moves down

$(3,2)$

Depth = 3.

Battery used = 3

Possible moves.

- Left
- Right
- Down

Level 4:-

Robot moves Right.

$(3,3)$

- Down.